

# Iris and Anterior Chamber Angle Neovascularization after Iodine 125 Brachytherapy for Uveal Melanoma

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**Purpose:** Iris neovascularization (INV) and anterior chamber angle neovascularization after radiotherapy for uveal melanoma may lead to neovascular glaucoma and enucleation. However, neovascularization of the anterior ocular segment may respond favorably to treatment with panretinal photocoagulation. The purpose of this study was to evaluate the frequency, interval to development, and predisposing factors of anterior ocular segment neovascularization following iodine 125 (<sup>125</sup>I) brachytherapy for uveal melanoma.

**Design:** Retrospective, interventional, consecutive case series.

**Participants:** Sixty-five patients (65 eyes), consecutively treated with <sup>125</sup>I brachytherapy for uveal melanoma from 1995 through 2000 and followed up after radiation therapy for 24 months or more.

**Methods:** Clinical findings and ultrasonography characteristics as well as treatment parameters were analyzed.

**Main Outcome Measures:** The frequency of INV was determined and the interval to development of INV as well as the predisposing factors were analyzed statistically.

**Results:** In 15 of 65 eyes (23%), INV was detected after <sup>125</sup>I brachytherapy at a mean  $\pm$  standard deviation of  $26.66 \pm 11.63$  months (median, 24 months; range, 9–48 months). Risk factors displaying the stronger correlation with INV were greater maximal tumor height ( $P < 0.01$ ), greater tumor vascularity ( $P < 0.01$ ), and disinsertion of horizontal rectus muscles ( $P = 0.01$ ).

**Conclusions:** After <sup>125</sup>I brachytherapy for choroidal melanoma, INV developed in 23% of eyes and was correlated with larger tumor size, greater tumor vascularity, and disinsertion of a horizontal rectus muscle. *Ophthalmology* 2005;112:505–510 © 2005 by the American Academy of Ophthalmology.

Radiotherapy, in the form of external beam (teletherapy) or radioactive plaque (brachytherapy), has proven effective for the local control of uveal melanoma.<sup>1,2</sup> However, radiotherapy may be associated with ocular complications, such as neovascularization of the retina, iris, and anterior chamber angle.<sup>1,3,4</sup> The development of the latter is often devastating, because it can lead to neovascular glaucoma (NVG), a common reason for performing enucleation at a later stage.<sup>3,4</sup> The prevention or early detection of the development of anterior ocular segment neovascularization, including iris neovascularization (INV) and anterior chamber angle neovascularization (ANV), is very important because it may be more effectively treated if recognized early.<sup>5</sup> Previous studies have identified risk factors associated with the development of INV after teletherapy for the treatment of

uveal melanoma.<sup>3,4</sup> In the case of brachytherapy, INV is well documented, although probably less common compared with teletherapy.<sup>1,6</sup> The present study examined risk factors associated with the development of anterior ocular segment neovascularization after iodine 125 (<sup>125</sup>I) brachytherapy for the treatment of uveal melanoma. Results could prove useful for the early detection of INV and ANV in patients treated with <sup>125</sup>I brachytherapy for uveal melanoma.

## Patients and Methods

The archived medical charts of patients with uveal melanoma treated consecutively with <sup>125</sup>I brachytherapy at the Ocular Oncology Center of Jules Stein Eye Institute during a 5-year period (1995–2000) were examined. Only patients with a minimum follow-up period of 24 months were included to allow for adequate postoperative observation period, as described elsewhere.<sup>4</sup> Furthermore, patients with recurrent tumors (having previously undergone other forms of treatment, including teletherapy) were excluded to rule out changes not directly related to brachytherapy. The protocol was approved by the Institutional Review Board in conformity with the tenets of the Declaration of Helsinki.

On presentation, no patient was diagnosed with INV or ANV. Furthermore, no patient was using any topical medications. In all cases, the diagnosis of uveal melanoma was based on history, fundus photography, indirect ophthalmoscopy, transillumination,

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